



# LIGHT CUM ROOM FRESHENER Module With Two Options

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**T**his light cum room freshener module offers two choices. In the first choice, the room freshener spray and a small LED light get activated for three to five seconds. In the second choice, where a switch is employed, the spray stops (after three to five seconds), but the LED continues to glow. The second choice can double up as a night bedside lamp cum room freshener with your favourite choice of scent.

Fig. 1 shows the authors' prototype of the entire setup. Fig. 2 shows the block diagram of the device. Components used to build this module are listed in the Bill of Materials table.

The motor used in the proposed setup is a modified motor. The gear combination found in Home Miller's 250V motor is applied to a 4000RPM, 12V DC motor, and one round end effector is attached. The purpose is to activate the room freshener during one full rotation to the motor. Torque is the main reason for choosing the Miller's motor gear combination.

The popular NE555 timer IC used in the circuit consists of comparators, flip-flops, resistors, and transistors, all integrated onto a single chip. It is available in different packages, including DIP (dual in-line package) and surface mount versions, making it easy to use in various circuit designs.

Additionally, IRLZ44N, a specific model of a power MOSFET (metal-oxide-semiconductor field-effect

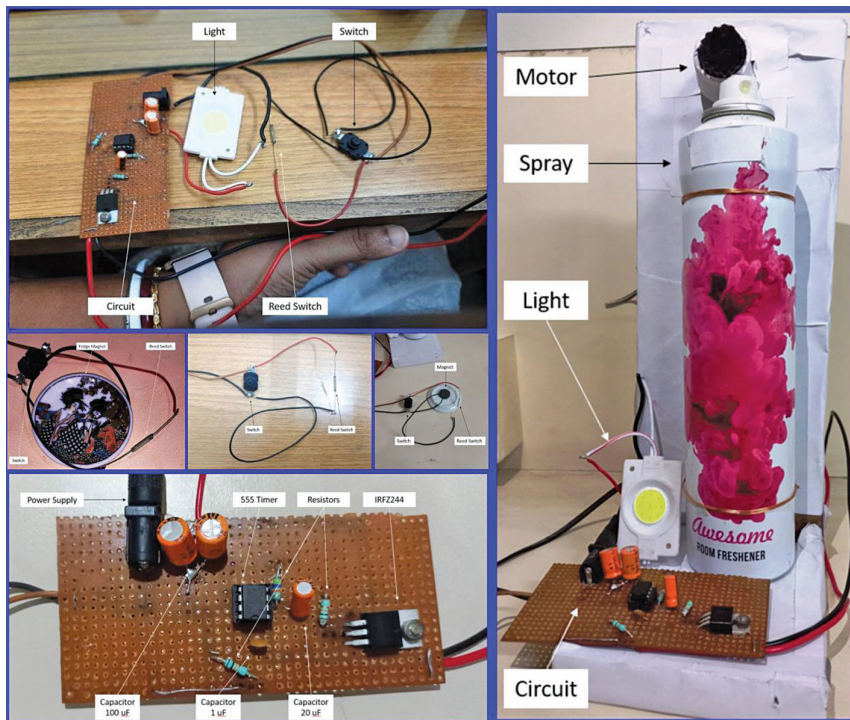


Fig. 1: Authors' prototype of the proposed set up of the light spray mechanism

transistor) commonly used in electronic circuits for switching applications, is employed. It is commonly used where high power switching is required, such as motor control, power supplies, inverters, and amplifiers.

A reed switch, also known as a reed contact, is used as a secondary switch. It is an electrical switch that operates

## BILL OF MATERIALS

| Components          | Quantity | Description                         |
|---------------------|----------|-------------------------------------|
| NE555 timer (IC1)   | 1        | IC that controls the on/off signals |
| IRLZ44N/IRFZ44 (T1) | 1        | Power MOSFET to drive motor         |
| Capacitors (C1-C4)  | 4        | For timing and ripple removing      |
| Resistors (R1-R3)   | 3        | For biasing                         |
| Modified motor (M1) | 1        | To activate the room freshener      |
| Switch SPDT (S2)    | 1        | To on and off the circuit           |
| Reed switch (S1)    | 1        | To power on or off using magnet     |
| 12V LED (LED1)      | 1        | To light up                         |
| 12V DC adaptor      | 1        | To power up the circuit             |

based on the magnetic field. It has two ferromagnetic reed blades enclosed in a glass tube filled with